Research Idea: Sombrero

ALMA Thesis Planner (automated pipeline)

A quasi-simultaneous millimeter/submillimeter SED of the Sombrero galaxy's nucleus: multi-band ALMA point-source photometry with a verified data floor, an informativeness forecast, and a single mid-semester narrative commitment

Using only the archival ALMA Cycle 0 dataset 2011.0.00754.S, the student measures the nuclear continuum flux density of the Sombrero galaxy (M104) in every spectral window actually delivered across Bands 3, 6, 7, and 9, characterizing the ~100–700 GHz spectrum of the nearest billion-solar-mass, extremely sub-Eddington AGN. The nucleus is a bright (~100 mJy), nearly uncontaminated point source, so photometry is done by fitting a point-source model per spectral window directly in the visibility (u–v) plane with standard CASA tools, cross-checked against image-plane measurements.

Week-one verification and forecast, before the plan is finalized. The student downloads the archive package and inventories: which of the 15 frequency settings and bands were executed; whether the package contains usable calibrated measurement sets or only raw ASDMs plus a legacy scriptForPI; and the date and time of every execution block. Crucially, this week also produces a one-page **informativeness forecast**: for each delivered band, using the actual spectral-window frequencies and separations, the student computes the expected spectral-index uncertainty $\sigma(\alpha)$ achievable within that band, assuming realistic residual spw-to-spw gain and bandpass errors of ~1–2% on top of thermal noise. This forecast determines *in advance* whether intra-band indices can discriminate anything (e.g., a flat compact-jet slope from a rising ADAF-type slope), or whether — given the small fractional bandwidth of a single Cycle 0 band — they serve only as consistency checks while the cross-band spectrum carries the science. The primary deliverable is chosen from evidence, not assumed.

Two decision gates, each resolved with the advisor in a scheduled meeting — not by the student alone:

(a) Calibration gate. If delivered calibrated products open cleanly, they are the baseline; recalibration via scriptForPI (possibly requiring a legacy CASA version) is a time-boxed improvement only. If delivered products are absent or broken and the time-boxed restore fails, the fallback is a calibration-and-recovery study of the dataset — an honest methods thesis, but a *different* thesis, so this branch activates only with explicit advisor and committee sign-off at the gate meeting; the student does not drift into it. In that branch, whatever subset of execution blocks calibrates successfully yields flux densities reported as measured data points — a solid thesis floor, without any claim that a single-band flux of a known ~100 mJy source constitutes a publishable result on its own.

(b) Epoch-and-coverage gate. The execution-block times and the informativeness forecast jointly fix the science framing. If the bands were observed quasi-simultaneously (within about a day), the goal is the full cross-band spectral-shape analysis: power-law, broken power-law, and curved power-law

fits with model-selection statistics, testing for the submm upswing or peak/turnover frequency the proposal targets. If bands are separated by days to weeks — where single-epoch-per-band data make curvature formally degenerate with source variability — the deliverable becomes whichever of two quantities the forecast shows is actually constraining: per-epoch intra-band spectral indices (variability-immune, used only if the forecast $\sigma(\alpha)$ is small enough to discriminate the jet vs. ADAF slopes), and the inter-epoch flux differences themselves, reported as a variability measurement or upper limit for the Sombrero's mm core, with cross-band curvature stated only as conditional ranges. Partial band coverage feeds into the same framework, with limits on where a spectral turnover could lie.

Complexity control. The gates all close by week six, and the thesis commits to *one* narrative at a mid-semester checkpoint — SED-shape thesis, variability-plus-indices thesis, or (with sign-off) recovery thesis. From that point the contingency machinery disappears from the write-up: the final document tells one story and relegates the decision process to a short methods subsection, so the thesis is about the black hole, not about its own decision tree.

The error budget separates thermal noise from the fits; absolute flux-scale systematics at the standard observatory-quoted Cycle 0 band-dependent values (~5–10%), with the dataset's own calibrators used as an internal consistency check rather than an independent derivation; and, in any multi-epoch branch, measured inter-epoch flux differences propagated directly. Fits treat within-band flux-scale errors as correlated. The interpretation confronts whichever measurements survive the gates with the generic predictions of the two accretion pictures the proposal invokes — a compact-jet power law versus an ADAF-type submm bump — with an ambiguous or limits-only outcome explicitly framed as a valid result. The scope remains one small (sub-hour) archival dataset on a bright point source, no new observations and no radiative-transfer modeling, completable in a semester with the writing phase protected regardless of what the archive contains.